



APEX ANALYTICA
MANIFOLD

CAUSAL-GRAPH FRAGILITY ANALYTICS

Manifold for Insurance & Reinsurance

A causal-inference research terminal for catastrophe modeling, treaty design, and cascade analysis across the reinsurance tower.

CONFIDENTIAL

PREPARED FOR AON

2026-05-08

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THE GAP

Where the current toolset stops.

- **CAT models fit historical event sets.** When the climate regime changes, the fit breaks. The model can describe the old regime, not the new one.
- **Cross-peril correlation is inferred from thin data.** The calibration fails exactly under stress, when coupling matters most.
- **Sanctions and treaty cascade live in scenario libraries.** They become additive bumps to an exposure number that doesn't know the underlying structure exists.
- **Stress tests give magnitude, not propagation.** When a primary fails, every treaty downstream doesn't fail at the same severity. The structure decides which links carry the cascade, and the structure isn't in the model.

Stress tests show how big. Causal graphs show how it propagates.

Most risk frameworks have the first and skip the second. Manifold builds the second on top of the structure your existing models assume away. The team comes from inside financial modelling, not adjacent to it: ex-HSBC AI, reinsurance, JHU systems engineering.

THE PLATFORM

Four engines, one configurable platform.

SPIRITES · CAUSAL DISCOVERY

Infers causal structure from claim, exposure, and balance-sheet panels. PCMCI on lagged returns and exposure deltas surfaces dependencies the historical fit missed.

TARSKI · FORMAL VERIFICATION

Verifies treaty terms, regulatory constraints, and ILS / parametric trigger structures against the inferred graph. Red-lines connections that violate the stated conditions.

PEARL · COUNTERFACTUAL INTERVENTION

Do-calculus on the graph. Answers the question stress tests gesture at and don't resolve: if you cut this exposure, what does the cascade look like.

PARETO · CASCADE SIMULATION

Per-node criticality estimators (BOCPD, persistent homology, transfer entropy) and full cascade simulation across the tower under stress.

THE SCORING FRAMEWORK

ΩF. Five pillars, per-node fragility.

Every node in the graph (counterparty, peril, treaty, jurisdiction, line of business) gets a score on five orthogonal pillars. The composite ΩF runs with bootstrap confidence intervals, so the readout shows uncertainty alongside the point estimate.

I

IRREPLACEABILITY

Substitutability of a counterparty, facility, or treaty under a freeze. Common-collateral choke points.

R

**RESTORATION
LATENCY**

Time to replace capacity after catastrophic failure. Recapitalisation speed under stress.

J

**JURISDICTIONAL
HAZARD**

Sanctions cascade, regulatory regime shifts, currency-zone fragmentation, cross-border CCP fragility.

C

CASCADE LOAD

Downstream exposure depth through the reinsurance tower. Common-collateral fire-sale loops.

T

TAIL DEPTH

Distributional depth beyond VaR (99.9% / EVT regime). Correlation-collapse asymmetry.

WHERE IT LANDS AT AON

Five concrete use cases across reinsurance and capital advisory.

- **CAT model augmentation.** Surface the structural correlations the AIR, RMS, or KCC fit missed: common-peril infrastructure dependencies, climate-driven shifts in coupling, sub-portfolio cascade depth.
- **Treaty cascade analysis.** If a primary insurer or a key reinsurance partner fails, what happens across the tower. Pearl do-calculus produces the quantified cascade with per-node fragility readouts.
- **Climate non-stationarity stress testing.** Explicit causal model of climate-loss linkages with regime-shift detection on the time series. Produces a defensible disclosure narrative for TCFD, NAIC, or Solvency II.
- **ILS and parametric trigger structuring.** Tarski formally verifies trigger conditions against the graph. Quantifies basis risk with structural causality, not statistical correlation.
- **Counterparty cascade across reinsurance partners.** Ω F per partner ranks fragility under stress. Interdiction analysis quantifies the marginal benefit of hardening any single link.

WHAT A PILOT DELIVERS

Twelve weeks, one question, four phases.

A pilot scopes Manifold to one question your team is currently facing: a treaty cascade, a peril correlation review, a counterparty cascade across reinsurance partners, or an ILS trigger portfolio. The engagement runs in four phases.

PHASE 1 · WEEKS 1-2

Data intake and graph reconstruction. Your team hands over the data shape. We map it to the Manifold schema and produce a first-pass causal graph. Short feedback loop with whoever knows your portfolio.

PHASE 2 · WEEKS 3-6

Per-node fragility readouts. Every node in the graph gets scored across the five Ω F pillars, with bootstrap confidence intervals. Top-3 fragility nodes ranked, with the cascade scenario driving each ranking.

PHASE 3 · WEEKS 7-9

Interdiction analysis. Pearl do-calculus quantifies the effect of each candidate intervention. If you cut this exposure, harden this counterparty, restructure this treaty: how does the system-level fragility move, and by how much.

PHASE 4 · WEEKS 10-12

Written deliverable and workshop. 40–60 page report, executive summary, and an on-site or remote walkthrough with your team. Configured Manifold instance left running for AON read-access at the end of the engagement.

WHY APEX

A team built inside the discipline, not adjacent to it.

CEO & PRINCIPAL SCIENTIST

Junaid Ghauri

D.Eng. Johns Hopkins. Bayesian modelling, COVID-19 analytics, reinsurance risk. Ex-CTO MARK LABS. Authored the Ω -Robustness foundations in his dissertation.

SOFTWARE & RESEARCH

Daniel Mastropietro

20+ years across energy, telecom, finance, government, manufacturing. Reinforcement learning, optimization, model interpretability. Doctoral candidate INP Toulouse. Fulbright alumnus.

HEAD OF RESEARCH

Georgios Korpas

Ph.D. Mathematics, Trinity College Dublin (visiting Stanford). Ex-HSBC AI and optimization. Quantum algorithms for collateral and continuous optimization.

RISK MODELING ADVISOR

Dr. Gonzalo L. Pita

Associate Research Scientist, JHU. Natural disaster risk, climate change, infrastructure vulnerability. Former Senior Consultant at the World Bank.

AI RESEARCH SCIENTIST

Jeremy Kulcsar

Ex-HSBC AI Research. Causality, portfolio optimization, deep learning. Author of the Ω F five-pillar scoring framework. Ecole Centrale Paris MSc CS.

SYSTEMS ENGINEERING ADVISOR

Dr. Takeru Igusa

Professor, Johns Hopkins. Systems science, applied math, epidemiology, resilience. NIH and CDC funded across 25 years of risk-modelling research.

HOW TO START

Four engagement shapes, sized to the question.

ΩF MINI-AUDIT

\$1,500

Flat • 5 business days

Send a sanitised AON dataset. Get a 12-page ΩF readout, top-3 fragility nodes, suggested interdictions, and a 30-min walkthrough.

CAUSAL RISK
ASSESSMENT

\$7.5-15k

2-3 weeks • scoped

Custom analysis on a specific question, such as a treaty cascade, a peril cluster, or a counterparty network. Configured Manifold instance plus written deliverable.

ENTERPRISE PILOT

\$150k+

3-6 months • embedded

Insurance-domain configuration of Manifold against AON's data shape. On-demand engineering, dedicated solutions support, contractual SLA.

PLATFORM LICENSE

From \$48k

Per seat / year

Continuous Manifold access for an AON team. Multi-domain seats with cross-asset bridges, live feeds, and the full estimator suite.

WHAT WE'RE ASKING

A walkthrough, then one of three follow-on paths.

Pick a use case AON cares about and we'll bring Manifold live on a 30-minute call, configured for that use case, and walk through one cascade scenario relevant to your book or capital-advisory work.

- **Path 1. Ω F Mini-Audit on a sanitised AON dataset.** \$1,500 flat. Concrete output in 5 business days.
- **Path 2. Scoped Causal Risk Assessment.** Pick one question; we deliver a written analysis in 2–3 weeks at \$7.5–15k.
- **Path 3. Enterprise Pilot.** 12-week embedded engagement, configured Manifold instance, \$150k+. The path forward if the walkthrough earns it.

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